

Lem ideal total

Lema 1. *Sea $I \subset R$ un ideal de un anillo R , entonces*

$$1 \in I \vee \exists a \in I \text{ invertible} \implies I = R.$$

Demostración: Veamos ambos casos por separado.

- I. Si $1 \in I$, por la propiedad de absorción, $\forall r \in R : r = r \cdot 1 \in I$, luego $I = R$.
- II. Si $a \in I$ es invertible, entonces $\exists a^{-1} \in R : aa^{-1} = 1$. Entonces, por la propiedad de absorción, tenemos que $1 \in I$ y, por el caso ??, esto implica que $I = R$. ■

Referenciado en

- `lem-cuerpo-iff-ideales-triviales`

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